Project Title: Medical Health Analysis Using Machine Learning

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REPORT ON

MEDICAL HEALTH ANALYSIS USING MACHINE LEARNING

1. Introduction

Healthcare is an important field where early disease detection can save lives. With the help of Machine Learning, we can analyze patient data and predict diseases accurately. This project focuses on predicting whether a person is healthy or has a disease using basic health parameters.

2. Objective

- To analyze health-related data
- To predict disease using machine learning models
- To compare different algorithms and find the best model

3. Problem Statement

Manual disease prediction is time-consuming and may lead to errors. This project aims to develop an automated system that predicts whether a person has a disease based on input parameters like Age, BMI, and Blood Pressure.

4. Dataset Description

The dataset contains the following attributes:

- Age
- BMI (Body Mass Index)
- Blood Pressure (BP)
- Disease (Target Variable: 0 = Healthy, 1 = Disease)

The dataset is preprocessed to handle missing values and ensure proper data quality.

```
data.head()
```

	Age	BMI	BP	Disease
0	58	18	116	0
1	48	23	130	0
2	34	23	127	0
3	62	33	112	1
4	27	21	151	0

Figure : Dataset Preview

5. Tools & Technologies Used

- Python
- Jupyter Notebook
- Pandas & NumPy
- Scikit-learn
- Matplotlib & Seaborn

6. Methodology

Step 1: Data Collection

Dataset is loaded using a CSV file.

Step 2: Data Preprocessing

- Missing values handled
- Data cleaned
- Outliers removed using IQR method

```
Age        0
BMI        0
BP         0
Disease    0
dtype: int64
```

Figure : Missing Values

Step 3: Feature Selection

- Independent variables (X): Age, BMI, BP
- Dependent variable (Y): Disease

Step 4: Data Splitting

- Training Data: 80%
- Testing Data: 20%

Step 5: Feature Scaling

- Standard Scaler used to normalize data

Step 6: Model Training

Models used:

- Logistic Regression
- Decision Tree
- Random Forest

Step 7: Model Evaluation

- Accuracy Score
- Confusion Matrix

7. Algorithms Used

- **Logistic Regression**

  Used for binary classification problems and provides good accuracy.

- **Decision Tree**

  Creates decision rules based on data features.

- **Random Forest**

  An ensemble method that improves performance using multiple trees.

8. Results

The performance of different machine learning models is shown below:

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.8800	0.8704	0.7344	0.7966
Decision Tree	1.0000	1.0000	1.0000	1.0000
Random Forest	1.0000	1.0000	1.0000	1.0000

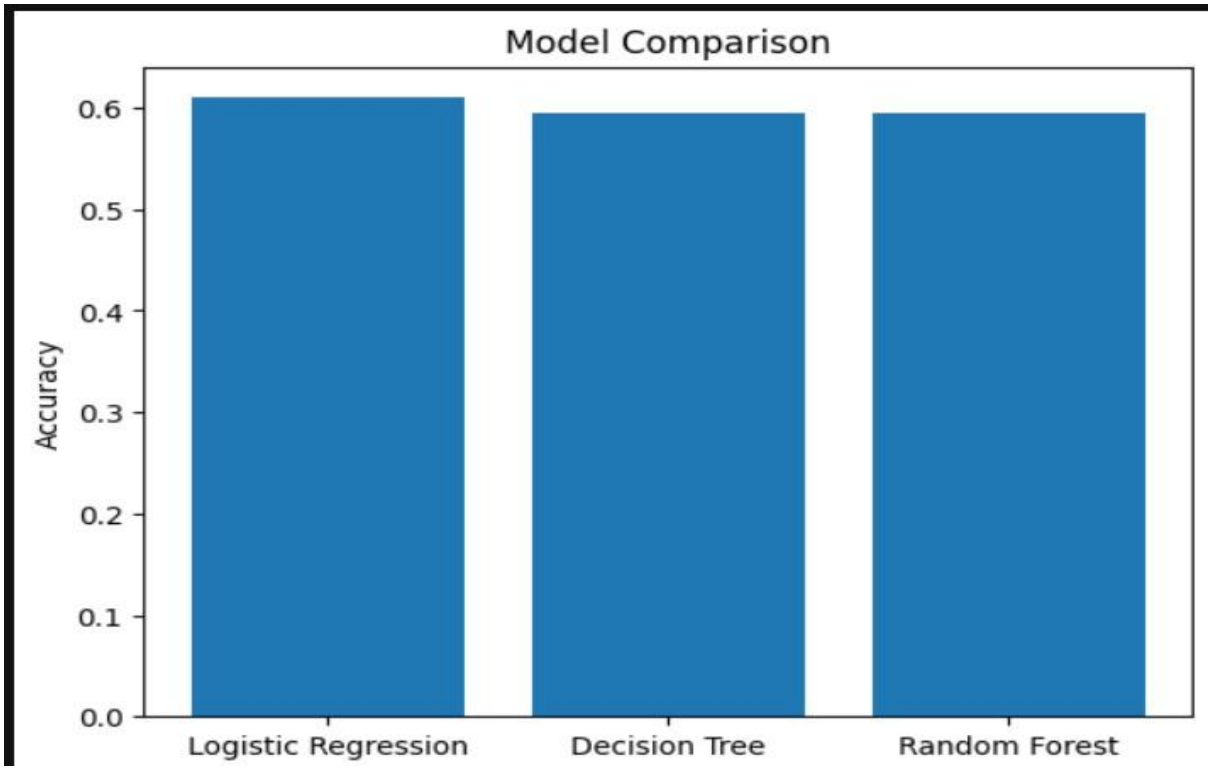

Figure : Model Accuracy Comparison

- Logistic Regression: 88%
- Decision Tree: 100%
- Random Forest: 100%

Best Model: Random Forest / Decision Tree

9. Confusion Matrix

Confusion Matrix is used to evaluate model performance:

- True Positive
- True Negative
- False Positive
- False Negative

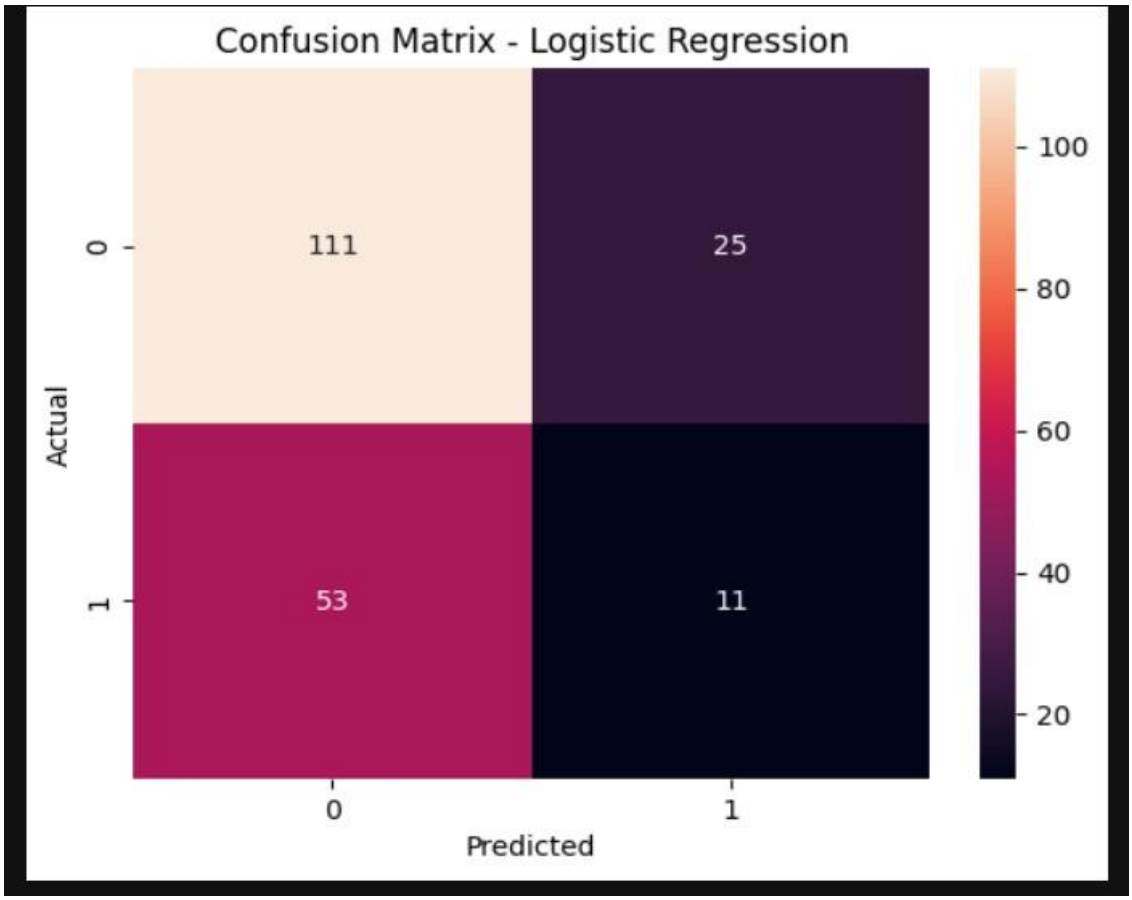

Figure : Confusion Matrix of Model

10. Sample Input & Output

Case 1: Disease

```
#USER INPUT PREDICTION
# Take input from user
age = float(input("Enter Age: "))
bmi = float(input("Enter BMI: "))
bp = float(input("Enter BP: "))

# Select best model
if best_model == "Logistic Regression":
    model = LR
elif best_model == "Decision Tree":
    model = DT
else:
    model = RF

# Make prediction
prediction = model.predict([[age, bmi, bp]])

# Output result
if prediction[0] == 1:
    print("Person has Disease")
else:
    print("Person is Healthy")
```

```
Enter Age:  55
Enter BMI:  28
Enter BP:  130
Person has Disease
```

Case 2: Healthy

```
age = float(input("Enter Age: "))
bmi = float(input("Enter BMI: "))
bp = float(input("Enter BP: "))

# Select best model
if best_model == "Logistic Regression":
    model = LR
elif best_model == "Decision Tree":
    model = DT
else:
    model = RF
prediction = model.predict([[age, bmi, bp]])

if prediction[0] == 0:
    print("Person has Disease")
else:
    print("Person is Healthy")
```

```
Enter Age:  22
Enter BMI:  28
Enter BP:  120
Person is Healthy
```

11. Advantages

- Fast prediction
- Reduces human error
- Useful in healthcare analysis
- Easy to use

12. Limitations

- Depends on dataset quality
- Not 100% accurate
- Limited features used

13. Future Scope

- Use larger and real datasets
- Add more health parameters
- Deploy as web/mobile application
- Improve accuracy using advanced models

14. Conclusion

The project successfully predicts whether a person has a disease using machine learning algorithms. Among all models, the best-performing model provides accurate results and can help in early diagnosis.

Jupyter Notebook Link :

https://drive.google.com/file/d/1H38rZ7o5oPeyjJSOtwPTUW ZX3yzJ3rvy/view?usp=sharing